Arvind Kumar Yadav

gs25mtech11203@iith.ac.in | yadavarvind40302@gmail.com | +91 7068203385 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.69
JAIRAM K I C KRANAIPUR PRAYAGRAJ	Class XII (Board of Highschool and Intermediate Education Uttar Pradesh)	2019	59.40%
ADRASH I C GHEENPUR ALLAHABAD	Class X (Board of Highschool and Intermediate Education Uttar Pradesh)	2017	85.16%

PROJECTS

Dynamically Optimal Battery Charging Protocols across Charging Cycles using Machine Learning

Master's Thesis | *IIT Hyderabad* | *May 2026-Present*

- **Investigated** battery ageing under repeated charging cycles, highlighting how fixed charging protocols become suboptimal as battery health deteriorates.
- **Analyzed** 357 research papers using an LLM-assisted systematic literature analysis framework to map AI-based battery modelling, charging optimization, and identify key research gaps in ageing-aware charging.
- **Developing** an ageing-aware battery state-transition model using **LSTM** and an adaptive charging strategy using **Soft Actor-Critic (SAC)** to dynamically optimize charging current across battery life.
- **Aiming to enable** safe, efficient, and degradation-aware charging by integrating battery-state prediction with **optimization-based safety constraints**, with the expected outcome of reduced degradation and improved battery lifetime.

Explainable Short-Horizon NO Forecasting in Wastewater Treatment | Python, Pandas, DuckDB/SQL, Scikit-learn

- **Developed** a one-hour-ahead NO forecasting pipeline from high-frequency wastewater-process data, using quality-aware hourly aggregation, feature engineering, and chronological splitting to **prevent data leakage**.
- **Benchmarked** Persistence, Ridge, Random Forest, and HistGradientBoosting using **MAE, RMSE, and R^2**; selected HistGradientBoosting based on validation performance, achieving **0.0338 MAE, 0.0530 RMSE, and 0.7988 R^2** on the held-out test period.
- **Identified** key predictive drivers through **permutation importance and error analysis**, finding recent NO history as the strongest signal, with **DO setpoint, DO, nitrate, and influent flow** providing additional predictive information; revealed **higher prediction errors under elevated NO conditions**.

Bayesian Material Flow Analysis of E-waste in India | *Python, Bayesian MCMC, Probabilistic Modeling*

Data Science Analysis | *IIT Hyderabad* | *2026*

- **Developed** a Bayesian material-flow model for 2010-2025 synthetic e-waste data, estimating collection (C/G)(C/G) and recycling (R/C, R/G)(R/C,\ R/G) fractions with uncertainty.
- **Performed** 4-chain MCMC inference with **R 1.0053, ESS > 900 and zero divergences**, demonstrating stable posterior sampling and convergence.
- **Quantified** cumulative material flows as **3.93 Mt collected and 3.67 Mt recycled**, with **94% Bayesian credible intervals** for uncertainty-aware flow estimation.

CNN-Based E-Waste Image Classification | *Python, TensorFlow, Keras*

- **Developed** a CNN-based image classifier for automated **10-class e-waste categorization**, training on **2,400 images** with separate validation and test sets.
- **Implemented** image augmentation, pixel normalization, batch normalization, max pooling and dropout in TensorFlow/Keras to improve model generalization for e-waste image classification.

SKILLS

- **Programming & Data:** Python, SQL, NumPy, Pandas, Matplotlib, MS Excel
- **Data Analytics:** Data Cleaning, Exploratory Data Analysis, Data Visualization
- **Machine Learning:** Classification, Random Forest, SVR, Gaussian Process Regression
- **Deep Learning:** CNNs, RNNs, LSTMs, GRUs, TensorFlow, Keras
- **Optimization & Probabilistic Methods:** Bayesian Optimization, Bayesian Inference, BoTorch
- **Domain:** Battery Systems

RELEVANT COURSES

Machine Learning for Process Systems Engineering | Data Science Analysis (Audit) | Advanced Numerical Methods | Electrochemical Energy Storage Systems: Batteries, Fuel Cells and Supercapacitors (Additional, Ongoing) | Supply Chain Management and Circular Economy | Instrumentation for Efficient Recycling and Automation

ACHIEVEMENTS & RESEARCH ACTIVITIES

Global Sustainability Challenge - Stanford University

Selected for Phase-2 with a proposal on **Hybrid Microbial Fuel Cells for Indium Recovery from LCD E-waste**.

WORK EXPERIENCE

North Central Railway -- Civil Engineering Intern | Jun-Jul 2023

- Gained hands-on exposure to **engineering quality monitoring, materials testing, and site-level problem solving** during railway infrastructure construction.

POSITIONS OF RESPONSIBILITY

PG Mentor, SUNSHINE-IITH: Mentoring and supporting incoming postgraduate students by facilitating their transition to IIT Hyderabad and coordinating academic and campus-related guidance.

Member, Inner Wellness & Development Team -- Indian Knowledge Systems Students Chapter IITH: Conduct morning **yoga, pranayama, and meditation sessions** for students as part of the club's wellness activities